CLASS 8 · CBSE · GANITA PRAKASH PART 1 · SHORT NOTES

Chapter 5: Number Play

Topic-wise short notes with solved examples

Parity, multiples and remainders studied with algebra; why the divisibility tests for 3, 9 and 11 work; digital roots; and cryptarithms.

1 Parity

- Even numbers have the form $2n$ and odd numbers the form $2n + 1$. Negative numbers such as -4 are also even.
- even $\pm$ even $=$ even; odd $\pm$ odd $=$ even; even $\pm$ odd $=$ odd.
- $a + b$ and $a - b$ always have the **same parity**, because they differ by $2b$.
- So for any numbers a, b, c, d, all the expressions $a \pm b \pm c \pm d$ have the same parity. Changing one sign changes the value by an even number.
- For four **consecutive** numbers, $a \pm b \pm c \pm d$ is always even, because two of the numbers are odd and two are even.
- An expression is always even if 2 can be taken out as a factor: $4m + 2q = 2(2m + q)$.

Example 1.1 Which of these are even for all integer values of the letters: $6m - 4n$, $3g + 5h$, $x^2 + 2$, $13k - 5k$?

SOLUTION

$6m - 4n = 2(3m - 2n)$: always even.

$3g + 5h$: odd when $g = 1, h = 0$, so not always even.

$x^2 + 2$: odd when x is odd.

$13k - 5k = 8k$: always even.

$6m - 4n$ and $13k - 5k$

Example 1.2 Place $+$ or $-$ signs between 5, 6, 7, 8. Can the result be 3?

SOLUTION

$5 + 6 + 7 + 8 = 26$ is even, and every other choice of signs has the same parity.

No. Every result is even.

2 Sums of consecutive numbers

- Every odd number is a sum of two consecutive numbers: $2n + 1 = n + (n + 1)$.
- The sum of three consecutive numbers is 3 times the middle one: $(n - 1) + n + (n + 1) = 3n$.
- The sum of four consecutive numbers is $4n + 6$, which is even but never a multiple of 4.

Example 2.1 The sum of four consecutive numbers is 58. Find them.

SOLUTION

$n + (n+1) + (n+2) + (n+3) = 4n + 6 = 58$, so $n = 13$

13, 14, 15, 16

Example 2.2 The greatest of five consecutive numbers is p. Write their sum.

SOLUTION

The numbers are $p-4, p-3, p-2, p-1, p$.

$5p - 10$

3 Even numbers and multiples of 4

- An even number is either a multiple of 4 (form $4p$) or 2 more than a multiple of 4 (form $4p+2$).

Numbers added	Sum	Multiple of 4?
$4p$ and $4q$	$4(p+q)$	Yes
$4p+2$ and $4q+2$	$4(p+q+1)$	Yes
$4p$ and $4q+2$	$4(p+q)+2$	No

- This is just like parity: two of the same kind give a multiple of 4, two of different kinds do not.

4 Properties of divisibility

- If a divides M and a divides N, then a divides $M+N$ and $M-N$.
- If A is divisible by k, then every multiple of A is divisible by k.
- If A is divisible by k, then A is divisible by every factor of k.
- If A is divisible by k and by m, then A is divisible by the **LCM** of k and m (not always by $k \times m$).
- The reverse statements can fail: a multiple of 8 can be a sum of two numbers that are not multiples of 8 ($72 = 50 + 22$); a multiple of 7 need not be a multiple of 28.

Example 4.1 Always, sometimes or never true: "If a number is divisible by both 6 and 4, it is divisible by 24."

SOLUTION

The LCM of 6 and 4 is 12. So the number is certainly divisible by 12.

12 is divisible by 6 and 4 but not by 24. 48 is divisible by all three.

Sometimes true

Example 4.2 Always, sometimes or never true: "The sum of a multiple of 6 and a multiple of 9 is a multiple of 3."

SOLUTION

$6x + 9y = 3(2x + 3y)$

Always true

Example 4.3 Always, sometimes or never true: "An odd number plus an even number is a multiple of 6."

SOLUTION

The sum is odd, but every multiple of 6 is even.

Never true

5 Remainders

- Numbers that leave remainder r on division by n have the form $nk + r$. Those leaving remainder 3 on division by 5 are $5k + 3$: $3, 8, 13, 18, \ldots$ The same numbers can be written $5k - 2$ with $k \geq 1$.
- **Remainders can be added, subtracted and multiplied.** If two numbers leave remainders r and s on division by n, their sum, difference and product leave the same remainders as $r + s$, $r - s$ and rs (reduced by multiples of n).
- If a number leaves the same remainder r with several divisors, it has the form (LCM of the divisors) $\times k + r$.

Example 5.1 On division by 7, the number 661 leaves remainder 3 and 4779 leaves remainder 5. Find the remainders of $4779 + 661$ and $4779 - 661$.

SOLUTION

Write $4779 = 7p + 5$ and $661 = 7q + 3$.

Sum $= 7(p + q) + 8 = 7(p + q + 1) + 1$. Difference $= 7(p - q) + 2$.

Remainders 1 and 2

Example 5.2 Find the smallest number that leaves remainder 2 on division by 3, remainder 3 on division by 4 and remainder 4 on division by 5.

SOLUTION

Each remainder is 1 less than the divisor, so the number is 1 less than a common multiple of 3, 4 and 5.

LCM$(3, 4, 5) = 60$

59

Example 5.3 Describe all numbers that leave remainder 2 on division by 3 and also on division by 4.

SOLUTION

Such a number is 2 more than a common multiple of 3 and 4.

$12n + 2$: **2, 14, 26, 38, ...**

6 Divisibility by 2, 4, 5, 8 and 10

- A number $\ldots dcba$ equals $\ldots + 1000d + 100c + 10b + a$. Every place value except the units is a multiple of 10 (and so of 2 and 5).

Divisor	Test
10	Units digit is 0
5	Units digit is 0 or 5
2	Units digit is even
4	Number formed by the last two digits is divisible by 4 (since 100 is a multiple of 4)
8	Number formed by the last three digits is divisible by 8 (since 1000 is a multiple of 8)

7 Divisibility by 9 and 3

- Each power of 10 is 1 more than a multiple of 9: $10 = 9 + 1$, $100 = 99 + 1$, $1000 = 999 + 1$.
- So a number and the **sum of its digits** leave the same remainder on division by 9 (and by 3).
- **A number is divisible by 9 exactly when its digit sum is divisible by 9, and by 3 exactly when its digit sum is divisible by 3.**
- Rearranging the digits does not change the digit sum, so it does not change divisibility by 9 or 3.

Example 7.1 Find the remainder when 7309 is divided by 9.

SOLUTION

$7 + 3 + 0 + 9 = 19$, then $1 + 9 = 10$, then $1 + 0 = 1$

1

Example 7.2 $31z5$ is a multiple of 9. Find the digit z.

SOLUTION

Digit sum $= 9 + z$ must be a multiple of 9.

$z = 0$ **or** $z = 9$

Example 7.3 How many multiples of 9 lie between 4300 and 4400?

SOLUTION

The first is 4302 (digit sum 9) and the last is 4392 (digit sum 18).

$$\frac{4392 - 4302}{9} + 1$$

11

8 Divisibility by 11

- Place values are alternately 1 more and 1 less than a multiple of 11: $1 = 0 + 1$, $10 = 11 - 1$, $100 = 99 + 1$, $1000 = 1001 - 1$.
- **Test**: starting from the units digit, put alternate $+$ and $-$ signs before the digits and add. The result has the same remainder as the number on division by 11. The number is divisible by 11 exactly when the result is 0 or a multiple of 11.
- If the result is negative, add 11 to get the remainder.

Example 8.1 Is 90904 divisible by 11?

SOLUTION

From the units digit: $4 - 0 + 9 - 0 + 9 = 22$

Yes, since 22 is a multiple of 11

Example 8.2 Find the remainder when 5529 is divided by 11.

SOLUTION

$9 - 2 + 5 - 5 = 7$

7

Example 8.3 Find the remainder when 328105 is divided by 11.

SOLUTION

$5 - 0 + 1 - 8 + 2 - 3 = -3$, and $-3 + 11 = 8$

8

9 Divisibility by other numbers

- To test divisibility by a product, use factors that have **no common factor**: 6 uses 2 and 3; 12 uses 3 and 4; 15 uses 3 and 5; 18 uses 2 and 9; 24 uses 3 and 8; 36 uses 4 and 9; 44 uses 4 and 11.
- Testing 24 with 4 and 6 fails, because the LCM of 4 and 6 is only 12. The number 12 passes both tests but is not divisible by 24.

Example 9.1 $48a23b$ is a multiple of 18. Find all pairs of digits (a, b).

SOLUTION

Divisible by 2: b is 0, 2, 4, 6 or 8.

Divisible by 9: $17 + a + b$ is a multiple of 9, so $a + b = 1$ or $a + b = 10$.

$(1, 0), (8, 2), (6, 4), (4, 6), (2, 8)$

Example 9.2 $3p7q8$ is divisible by 44. Find all pairs (p, q).

SOLUTION

Divisible by 4: $q8$ is a multiple of 4, so q is 0, 2, 4, 6 or 8.

Divisible by 11: $(8 + 7 + 3) - (p + q) = 18 - (p + q)$ is 0 or 11. Since $p + q = 18$ needs $q = 9$, we get $p + q = 7$.

$(7, 0), (5, 2), (3, 4), (1, 6)$

10 Digital roots

- Add the digits of a number, then the digits of the result, and so on until one digit is left. That digit is the **digital root**.
- The digital root is the remainder on division by 9, except that multiples of 9 have digital root 9.
- Digital roots of consecutive numbers run $1, 2, \ldots, 9, 1, 2, \ldots$ Adding 9 to a number does not change its digital root.
- Digital roots can check calculations: the digital root of a sum (or product) equals the digital root of the sum (or product) of the digital roots. This check was described by Āryabhaṭa II (about 950 CE).

Example 10.1 Find the digital root of 489710.

SOLUTION

$4 + 8 + 9 + 7 + 1 + 0 = 29$, $2 + 9 = 11$, $1 + 1 = 2$

2

Example 10.2 What is the digital root of $9a + 36b + 13$?

SOLUTION

$9a + 36b + 13 = 9(a + 4b + 1) + 4$

4

Example 10.3 Check $234 \times 56 = 13104$ using digital roots.

SOLUTION

Digital roots: 234 gives 9, 56 gives 2. Product of roots $= 18$, digital root 9.

13104 has digit sum 9.

The roots agree, so the answer passes the check.

11 Cryptarithms

- In a cryptarithm each letter stands for one digit, different letters stand for different digits, and a number never starts with 0.
- Useful clues: the units digits, carries, the number of digits in the answer, and parity.

Example 11.1 Solve AB $+$ 37 $=$ 6A.

SOLUTION

Units: $B + 7$ ends in A. Tens: $A + 3$ plus any carry is 6, so $A = 3$ with no carry or $A = 2$ with a carry.

If $A = 3$, $B + 7$ ends in 3 gives $B = 6$, which produces a carry: not allowed. If $A = 2$, $B + 7 = 12$ gives $B = 5$.

$25 + 37 = 62$

Example 11.2 Solve PQ $\times$ 8 $=$ RS.

SOLUTION

The product has two digits, so PQ is at most 12. PQ $= 10$ gives 80 (repeats 0), PQ $= 11$ repeats a digit.

$12 \times 8 = 96$

Example 11.3 Solve JK $\times$ 6 $=$ KKK.

SOLUTION

KKK $= 111\times$ K $= 37 \times 3\times$ K, so JK $\times 2 = 37\times$ K. K must be even.

K $= 2$ gives JK $= 37$ (units digit not 2). K $= 4$ gives JK $= 74$.

$74 \times 6 = 444$

12 Common mistakes

- Saying "divisible by 4 and 6 means divisible by 24". Use the LCM, which is 12.
- Adding digits to test divisibility by 11. That test uses **alternating** signs.
- Using the units digit to test divisibility by 3 or 9.
- Giving digital root 0 for a multiple of 9. It is 9.
- Deciding a statement is always true from a few examples. Use algebra, or find one counter-example.

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